Chuwen Zhang

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EDUCATION

Stony Brook University (SUNY) - New York, USA *Aug 2024 - May 2026*
B.S. in Applied Mathematics and Statistics; Minor in Finance GPA: 3.8/4.0
Relevant Course: Data Analysis, Data Mining, Statistical Methods for Data Science, Principles of Finance, Investment Analysis, Financial Management, Financial Accounting, Managerial Cost Analysis & Applications

Anhui University (AHU) - Hefei, China *Sep 2022 - Jun 2024*
Bachelor of Science in Applied Statistics GPA: 3.8/4.0
Relevant Course: Calculus, Probability Theory, Data Structure, Network Data Science: An Introduction, Mathematics Experiment Design

PUBLICATION

- Chuwen Zhang*, Xingchu Zhang, Rui Mao, Zehao Deng, *PAED: Physics-Anchored and Emotion Disentangled Memory Difusion for Talking Face Generation,* Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD 2026), DOI: https://doi.org/10.56028/aetr.15.1.1238.2025.
- He Yu*, Chuwen Zhang*, *Generating Missing Modalities: A Conditional Diffusion and Transformer Approach for Emotion Recognition*, Vol. 15 No. 1 (2025): 2025 International Conference on Artificial Intelligence, Mechanical Materials, Energy and Environment (AIMMEE 2025).
  DOI: https://doi.org/10.56028/aetr.15.1.1238.2025

INTERNSHIP

ZTE Corporation - Wuhan, China *May 2025 - Aug 2025*
AI Algorithm Engineer Intern | Research Center
- Designed and implemented a multi-stage agentic pipeline for automated error analysis and resolution using Router and Handoff architectures, decomposing complex issues into code risk analysis, system diagnostics, and solution synthesis
- Built intent recognition and query decomposition with specialized sub-agents, and developed context engineering with short- and long-term memory, integrating Vector Store – based RAG, knowledge base, and knowledge graph to enable accurate multi-hop reasoning over internal and external technical resources
- Implemented tool/MCP integration and LLM-as-Judge evaluation, enabling structured tool calling, robust error handling, and continuous optimization through prompt A/B testing

DP Tech - Hangzhou, China *Feb 2024 - May 2024*
AI Algorithm Engineer Intern
- Participated in the end-to-end training pipeline of core machine learning models, including data preprocessing, feature engineering, model tuning, and performance evaluation
- Conducted iterative model optimization through multiple training cycles, achieving an approximate 5% improvement in model accuracy
- Assisted in analyzing model performance and refining feature representations to enhance predictive capability

RESEARCH

Multimodal Data Completion with Prompt Learning and Diffusion Models *Apr 2025 - Dec 2025*
Leader | Supervisor: A/Prof. Yi Xu (AHU)
- Proposed a multimodal modality completion framework that integrates Conditional Diffusion with Transformer architectures to generate missing text, audio, and visual modalities for emotion recognition
- Introduced a diffusion-based generation module conditioned on modality availability, enabling high-fidelity completion under diverse missing-modality patterns
- Implemented end-to-end joint training with Transformer encoders and conducted controlled experiments to determine effective diffusion conditioning strategies and model configurations

Optimization of Emotional Consistency in Audio-Driven Video Generation *Dec 2024 - Mar 2025*
Leader | Supervisor: Prof. Björn W. Schuller (Imperial College London)
- Proposed a Physics-Anchored Emotion-Decoupled Diffusion model (PAED) based on Stable Diffusion and VAE frameworks, designing two core modules: a physics-guided Facial Action Unit (AU) momentum regulator and a cross-modal emotion disentangler
- Implemented the AU momentum regulator with weighted KL-divergence constraints to decouple auditory emotion from visual facial expressions, enabling joint training of the cross-modal disentangler

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- Validated model effectiveness through quantitative, qualitative, and ablation studies, achieving a 3.3% improvement in AU accuracy and reduced expression jitter in audio-driven video generation

Error Decomposition in Combined Forecasting Methods *Feb 2024 - Jun 2025*

Member | Supervisor: Prof. Huayou Chen (AHU)

- Developed a theoretical framework for error decomposition in weighted-average forecasting, analyzing the contributions of RMSE, bias, and variance
- Applied the decomposition method to evaluate the stability and accuracy across multiple models, providing methodology for ensemble forecasting

SELECTED PROJECT

Biomedical Informatics Bootcamp *Jan 2025 - Mar 2025*

- Built and evaluated core machine learning models including supervised and unsupervised methods (classification, regression, clustering, etc.), with emphasis on feature selection and cross-validation
- Studied RNA-seq informatics workflow and applied computational thinking to biological data analysis problems
- Studied modern NLP architectures including transformers and LLMs

Insurance Strategies and Community Planning in an Era of Extreme Weather *Jan 2024*

- Designed a dynamic evaluation model integrating confidence factor, SARIMA, PCA, and group decision-making
- Proposed strategies for insurance companies to achieve long-term stability under extreme weather conditions and guided community resilience planning

Optimal Design of Heliostat Field Layout *Sep 2023*

- Applied simulated annealing and cluster analysis to optimize heliostat elevation angles and layouts in tower solar thermal power plants

COMMUNITY ACTIVITY

Vice President & Co-Founder, AHU Mathematics Club *Oct 2024 - Present*

- Organized and led over 20 math discussions, seminars, and competitions with 1,000+ cumulative participants, fostering academic exchange and student interest in mathematics

Vice leader of Future Voyage Team, Summer Practice, AHU *Jul 2023*

- Conducted a field survey to analyze the current state of education in county and rural areas, as well as the educational challenges faced by left-behind children, and proposed targeted recommendations
- Recognized as an Outstanding Team and awarded Outstanding Individual

MISCELLANEOUS

- Technical skills: Python(PyTorch, Tensorflow), Java, SQL, R, Matlab, SPSS, Power BI, Git
- Awards: Dean’s List (Spring 2026, Fall 2025), First Prize Exchange Scholarship (2026), H Prize of MCM (Jan 2024), Provincial 2nd Prize of CMUCM (Sep 2023), AHU 3rd Prize Scholarship (Jun 2023), Provincial 3rd Prize of NCSSMC (Sep 2023)
- Languages: English (Fluent), Mandarin (Native), French (Basic)